School of Computing
Software Engineering Elective 1
SELECT1

1 COURSE IDENTIFICATION

COURSE CODE: SELECT1

COURSE TITLE: Software Engineering Elective 1

DESCRIPTION: In this course, we will learn Python development from basic scripting to building complex applications in Python. We will create modern day applications and learn how to solve programming challenges the Python way.

CREDIT UNITS: THREE (3) Units

2 LEARNING OUTCOMES

LO1

Understanding the fundamentals of programming in Python

LO2

Understanding the basic data processing and manipulation in Python using data structures built in the language

LO3

Developed critical thinking on how to solve various programming challenges written in Python.

LO4

Ability to produce complex applications using Python

3 COURSE OUTLINE

WEEK 1
- **Orientation**: Overview of the Course Syllabus and Setting of Expectations
- **Introduction to Python**
- **Hello, world!**
- **The Python documentation**
- **Values and Types**

WEEK 2
- **Variables**
- **Operators and operands**
- **Expressions and Statements**
- **Interactive and Script Mode**
- **Order of Operations**

WEEK 3
- **String operations**
  - **String functions**
  - **Indices**
- **Comments**
- **Built-in Modules**
- **Modulus**

WEEK 4
- **Boolean expressions**
- **Logical Operators**
- **Conditionals**
- **Built-in Functions**

WEEK 5
- **Custom Functions (intro)**
- **Parameters**
- **Function return values**
- **Recursion**

WEEK 6
- **Iterations**

WEEK 7
Midterm Examination

MIDTERM

WEEK 8
- **Lists**
- **Dictionaries**
- **Tuples**

WEEK 9
- **Package/Module Management**
- **PIP**
- **FIles**

WEEK 10
- **Python GUI**

WEEK 11
- **Classes and Objects**
- **Classes and Functions**
- **OOP Concepts in Python**

WEEK 12
- **OOP Concepts in Python**

WEEK 13
- Final Examination

FINAL

WEEK 14
- Machine Problem Deadline and Demo

4 ASSESSMENT TOOLS

- CLASS PARTICIPATION
- EXAMS
- SEATWORKS
- MACHINE PROBLEMS

5 GRADING SYSTEM

Midterm Exam	25%	Week 7
Final Exam	25%	Week 13
Machine Problem	20%	Week 14
Seatworks	20%	
Class Participation	10%	

6 REFERENCES

- Think Python by Allen Downey
- Head First Python by Paul Barry

7 INSTRUCTOR'S INFORMATION

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